The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 Axiomatic Metaphysics and the Causal Tracking Loop

The system trajectory of an asset is formalized as a higher-order relational tracking loop:

$$g(m, u, v) \equiv \exists y[R_{\text{system}}(m, u, y) \wedge E(m, u, y, v)] \tag{1}$$

where u represents environmental Navier-Stokes advection forces, m represents the external divine control input vector, y represents human or machine output behaviors, and E represents the absolute spiritual evaluation function.

2.1 Corollary 1.1: Anthropological Incompleteness and Vacuum Collapse

When the external control vector m is stripped from the network, human judgment operates as a strictly closed formal system, F. By Gödel's First Incompleteness Theorem, a closed system cannot generate its own semantic validation ($F \nvdash (v_{\text{human}} = v)$). Syntactic legalism remains blind to absolute truth, causing a critical valuation $v_k \in V'$ to become structurally unobtainable:

$$\exists v_k[V'_{v_k} \wedge \neg h(u, v_k)] \tag{2}$$

Without the active insertion of the divine control input vector, the spatiotemporal execution plane collapses into an empty relational set ($[\,]$)—a mathematical vacuum. Christ operates as the external Meta-Language that breaks this internal syntax boundary limitation.

3 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 37}$. For each node i, the vector decomposes into six distinct socio-pragmatic and deontic tensor blocks:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{3}$$

where $\boldsymbol{e}_i$ encapsulates the Barna loop splits and Pew macro-advection wavelets, $\boldsymbol{g}_i$ tracks continuous JAX coordinate steps, $\boldsymbol{p}_i$ and $\boldsymbol{r}_i$ pack the level-k PPL choice models and Rational Speech Act distributions, $\boldsymbol{s}_i$ maps the Searlean speech-act flags (X counts as Y in C), and $\boldsymbol{t}_i$ maintains the time-bounded legal compliance trajectories. The Graph Convolutional Network layer recursively convolutes these nodes over the empirical interaction topology:

$$h_i^{l+1} = \text{ReLU}\left(W^l \frac{1}{\sqrt{\deg_i}\sqrt{\deg_j}} \sum_{j \in \mathcal{N}_i} h_j^l \right) \tag{4}$$

This allows the network to smoothly compress giant graph spaces into a low-dimensional virtual 0-simplex token ready for total visible decision tree lookup routing.